

Groundwater

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- Exam corrections due now (in person)
- Final won't be cumulative
- HW6 deadline will be pushed to Nov 4

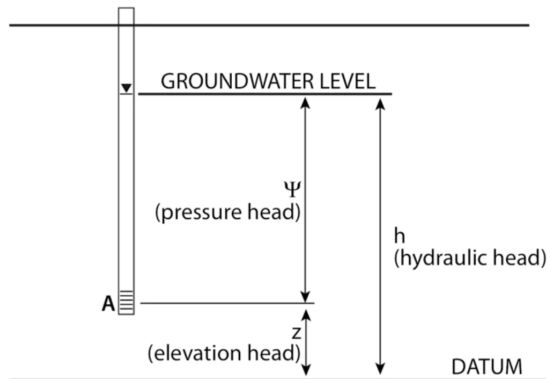
Learning Objectives

- 1 Aquifer Types and Properties
- 2 Groundwater Flow Equation(s)
- 3 Regional Groundwater Flow
- 4 GW/SW Interactions
- 5 Groundwater Balance Components
- 6 Groundwater Development

Hydraulic Head

$$h = z + \frac{p}{\rho_w \cdot g} = h_z + h_p$$

where z is elevation above a datum, p is pressure, ρ_w is density of water, and g is gravitational acceleration. Since ρ_w and g are constant, this can be simplified to h_p and the two terms are referred to as the "elevation head" and "pressure head."



Pressure Transducers

The pressure (at the base) exerted by a column of water is directly proportional to its height.

$$h_p = \frac{p}{\rho_w \cdot g} = \frac{p}{\gamma_w}$$

Pressure transducers record pressure (p) of overlying column of water AND column of air. So to get pressure head (h_p), you have to subtract atmospheric pressure from the total pressure measured by the transducer. How?



Groundwater Flow Equation

3-D, transient, anisotropic, heterogeneous. Derived from Darcy's Law and conservation of mass:

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x \left(\frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} K_y \left(\frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} K_z \left(\frac{\partial h}{\partial z} \right)$$

Homogeneous K

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x \left(\frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} K_y \left(\frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} K_z \left(\frac{\partial h}{\partial z} \right)$$

Isotropic Flow

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x \left(\frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} K_y \left(\frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} K_z \left(\frac{\partial h}{\partial z} \right)$$

Steady-state flow

$$S_s \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} K_x \left(\frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} K_y \left(\frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} K_z \left(\frac{\partial h}{\partial z} \right)$$

Laplace Equation

3-D, steady-state, homogeneous, isotropic flow is referred to as the **Laplace Equation**

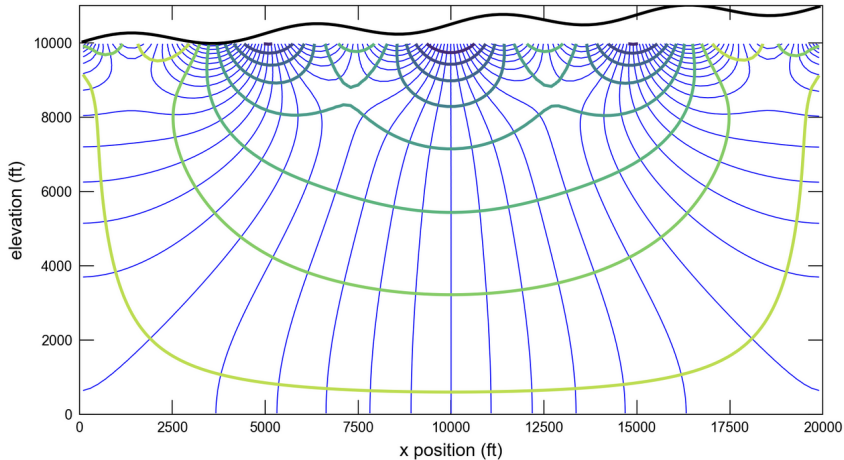
$$0 = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2}$$

Can be solved analytically in very simple situations, but is often solved numerically. Flow net construction uses a 2-D version of the Laplace equation in the x- and z-direction:

$$0 = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial z^2}$$

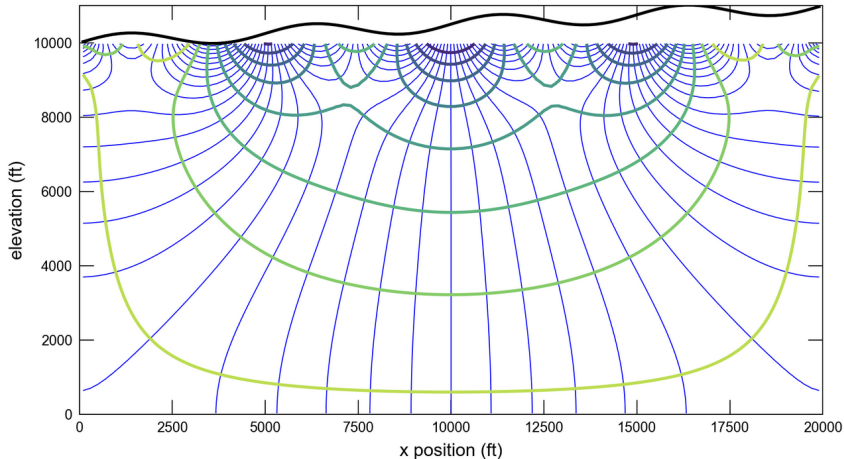
Flow Nets

Using the Laplace equation, assuming homogeneous K and assuming some simple boundary conditions, solve for head in the discretized grid.



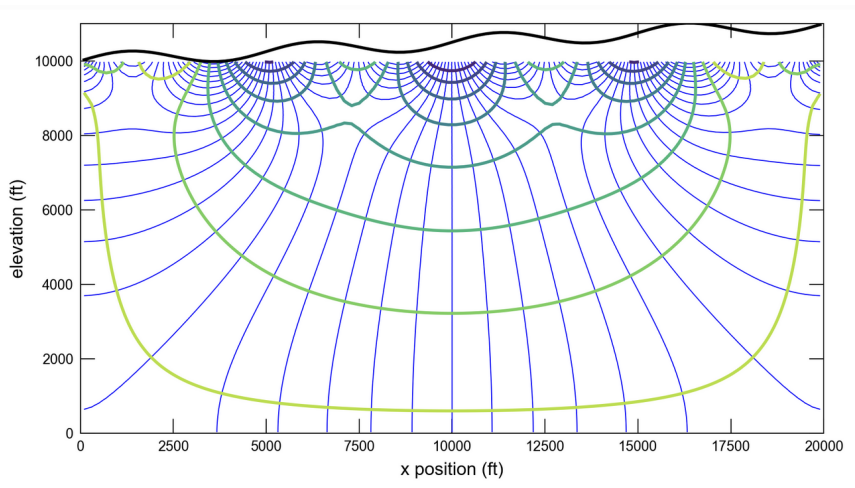
Flow Nets

Lines of equal head are contoured (**equipotential lines**). **Flow lines** (representing groundwater flow) are drawn to perpendicularly intersect equipotential lines.



Flow Nets

The area between flow lines is called a **flow tube**



Flow Net Boundary Conditions

- Base and sides are impermeable (no flow boundary): equipotential lines are perpendicular and flow lines are parallel.
- Specified (constant) head: equipotential lines are parallel and flow lines are perpendicular.
- Water table is a line of variable but known head. Equipotential lines are neither perpendicular nor parallel.

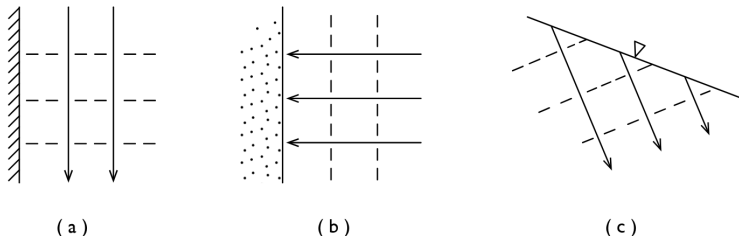
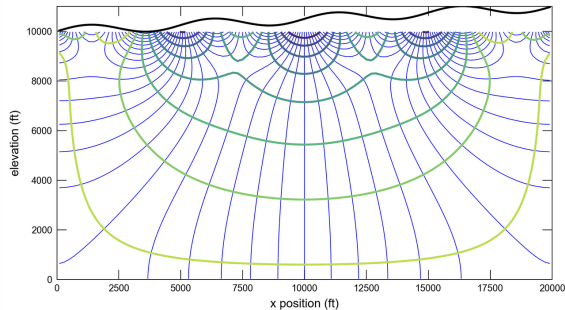


Figure 5.1 Groundwater flow in the vicinity of (a) an impermeable boundary, (b) a constant-head boundary, and (c) a water-table boundary.

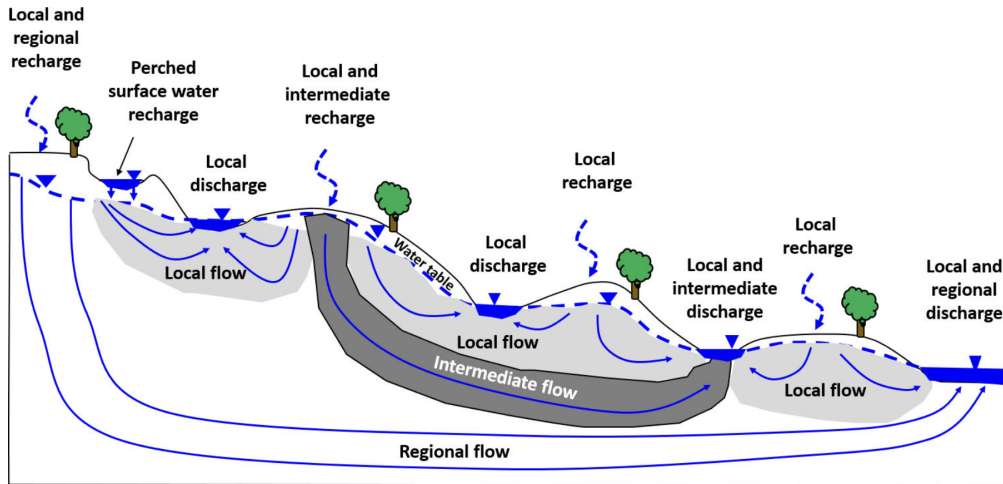
Flow Nets, cont'd

- If drawn to scale, flow tubes represent equal discharge.
- If homogeneous, the head distribution is invariant to hydraulic conductivity, and depends only on the boundary conditions.
- Hydraulic conductivity only comes into play when calculating discharge.
- Heterogeneous systems require numerical solution.



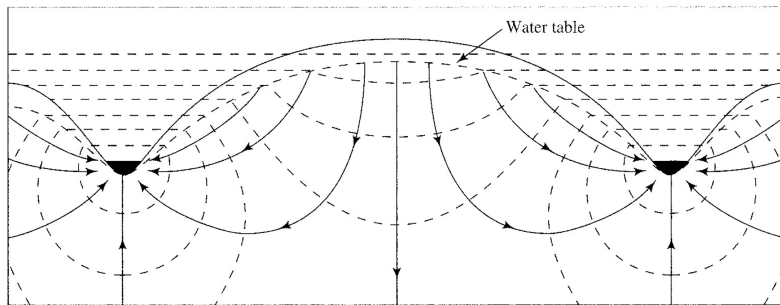
Regional Groundwater Flow

At the regional scale, Tóth (1963) showed that several flow systems are typically present in unconfined groundwater systems: local, intermediate, and regional.

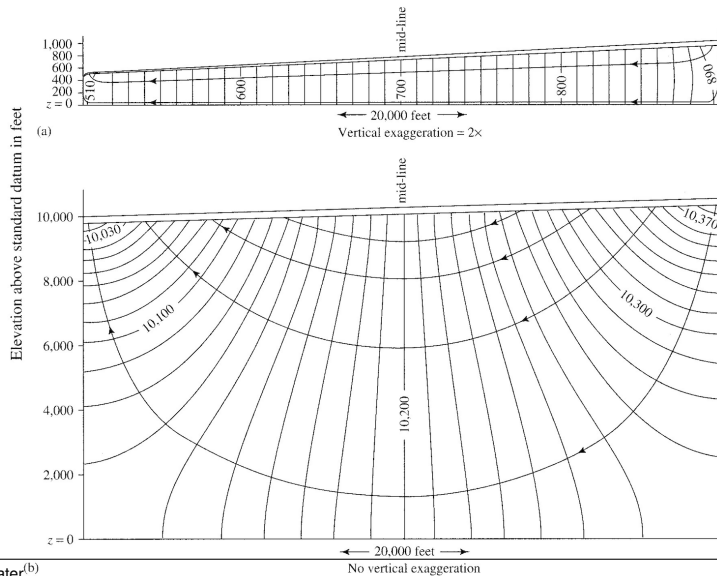


Features of Regional Flow

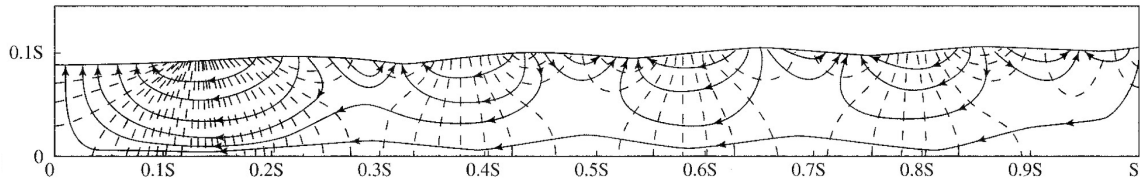
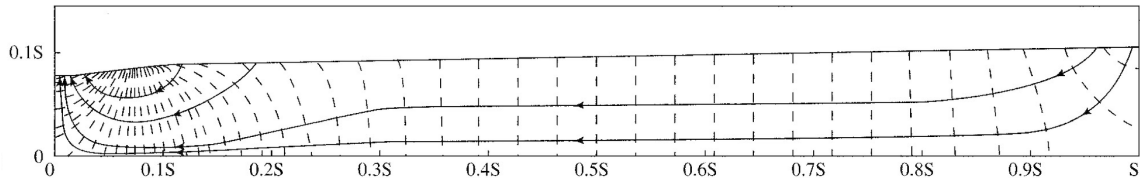
- Recharge is roughly balanced by discharge (in un-developed systems).
- **Recharge area:** portion of drainage basin where groundwater is directed away from the water table.
- **Discharge area:** portion of drainage basin where groundwater is directed towards the water table.



Effects of Aquifer Thickness

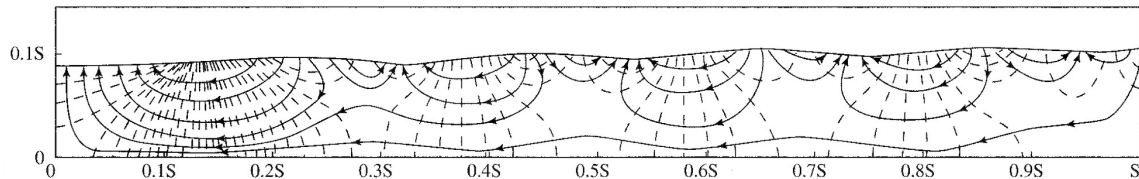
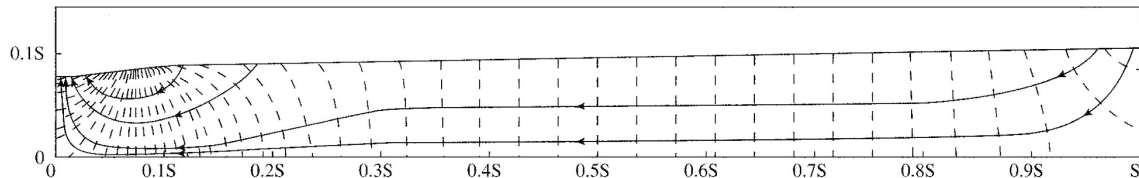


Effects of Topography



Dupuit-Forcheimer Assumption

For aquifer systems with a shallow slope and a length $\gg$ thickness, vertical flow can be negated. This is the **Dupuit-Forcheimer Assumption**.

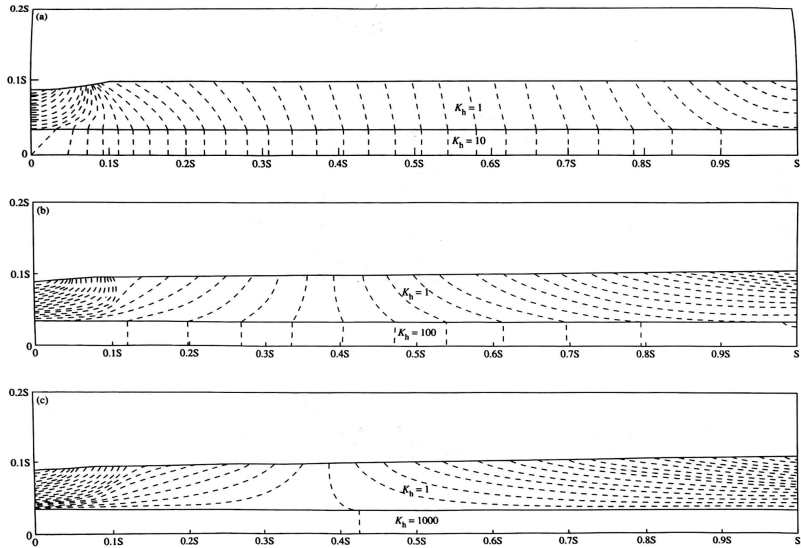


Effects of Geology

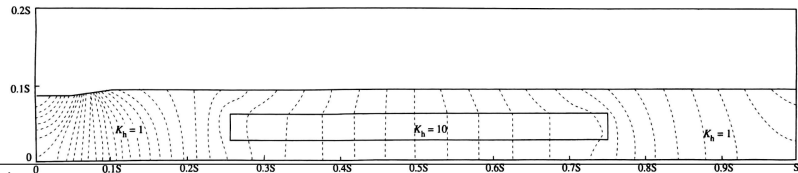
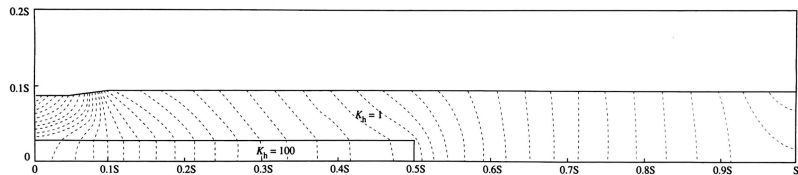
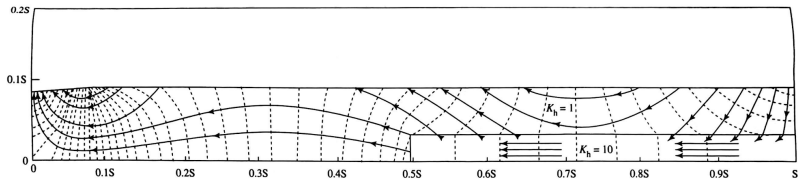
Geology exerts enormous control over the occurrence and flow of regional aquifer systems, generally fitting in one of three factors:

- 1 **Lithology:** (physical characteristics of rocks) control hydraulic conductivity distribution
- 2 **Stratigraphy:** geometric and age relations among various formations/lithologies. Typically layered.
- 3 **Structure:** the features and arrangements of rocks that result from forces like tectonic plate movement, including folds, faults, and joints.

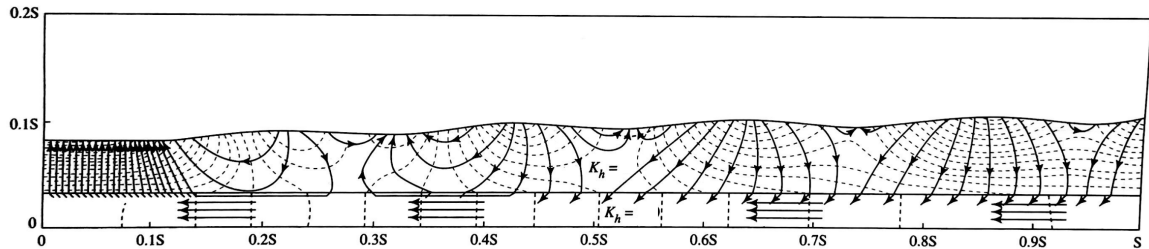
Effects of Geology



Effects of Geology

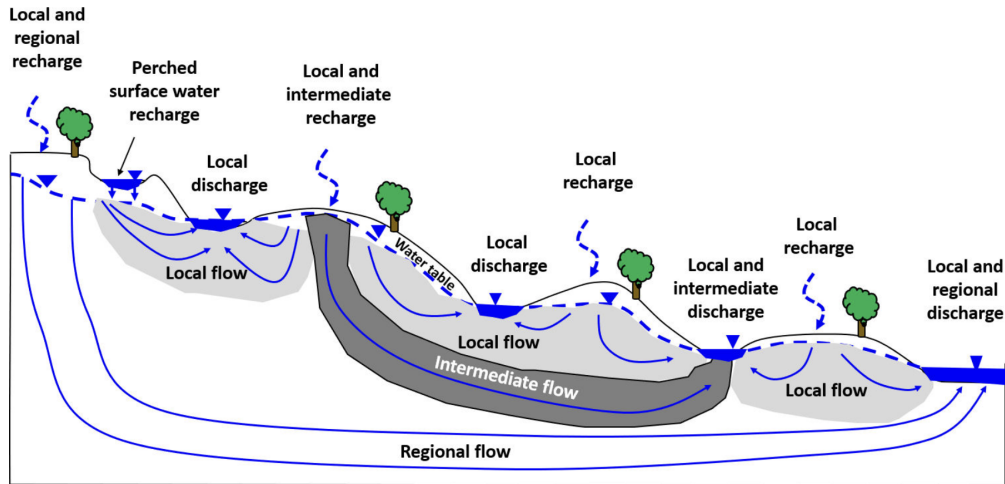


Effects of Geology



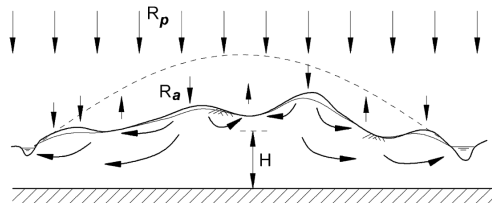
Are water tables a subdued replica of the land surface?

Tóth assumed that the water table generally follows the land surface, but this is not always the case.



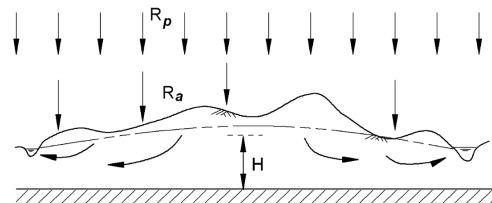
Depth and Form of Water Table

Haitjema & Mitchell-Bruker (2005) showed that the position and form of the water table depends on the dimensionless ratio of recharge (L/T) over hydraulic conductivity (L/T).



(a) $R_p \gg R_a$

k is low

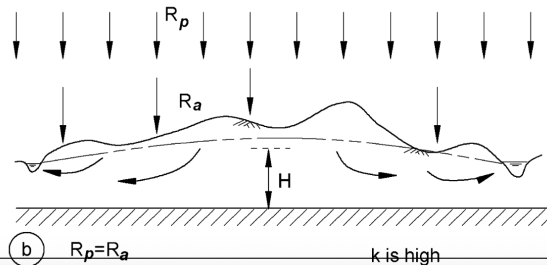
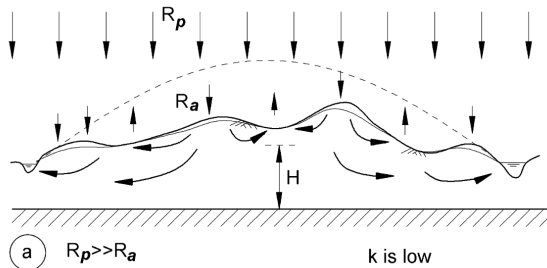


(b) $R_p = R_a$

k is high

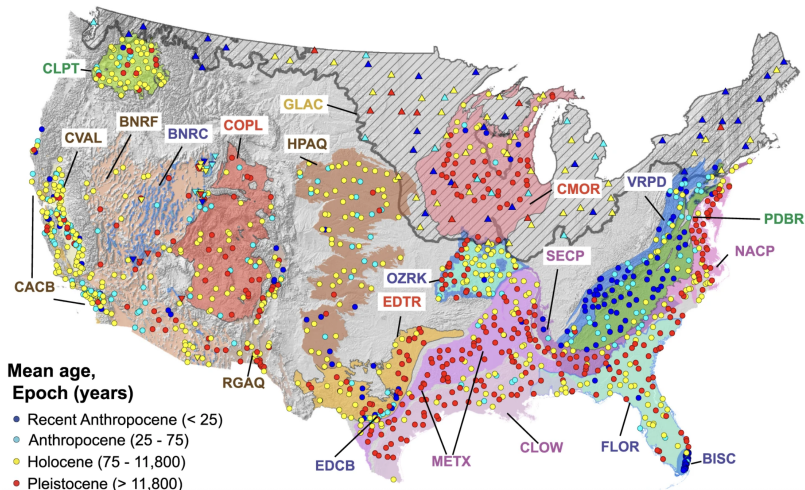
Depth and Form of Water Table

- High R /Low K produces a "topographically-controlled" water table
- Low R /High K produces a "recharge-controlled" water table.

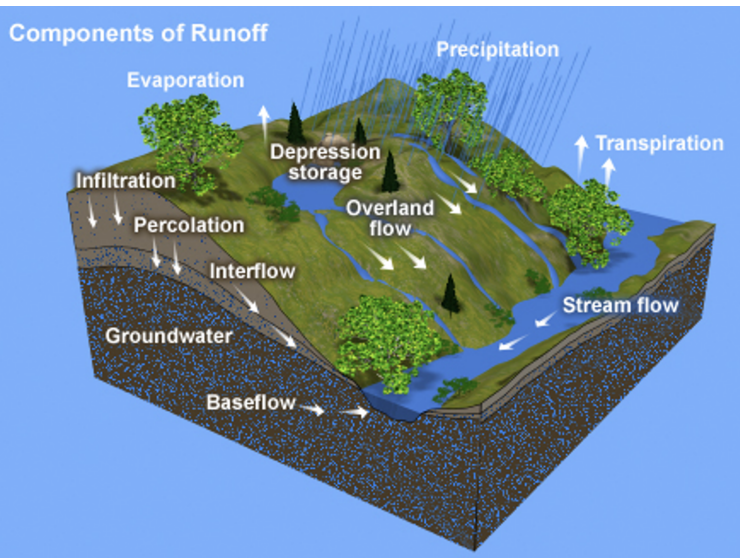


Implications

The position of the water table is critical for whether local/intermediate flow systems are present, with implications for water age and renewability/susceptibility to contamination.



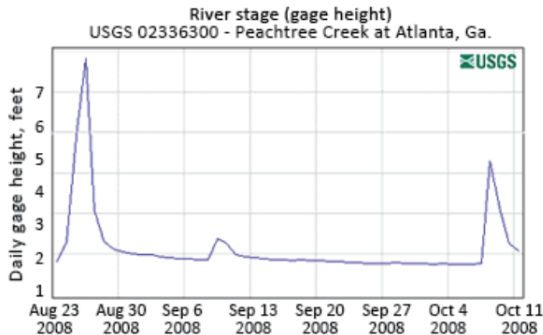
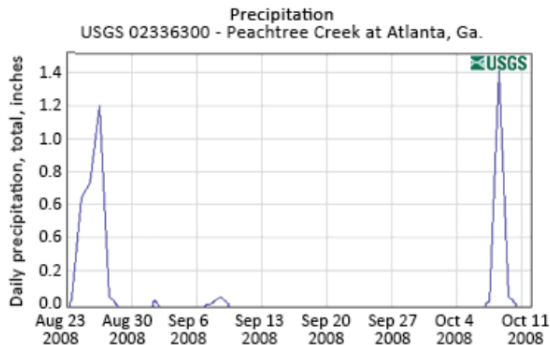
Runoff-Generation Mechanisms



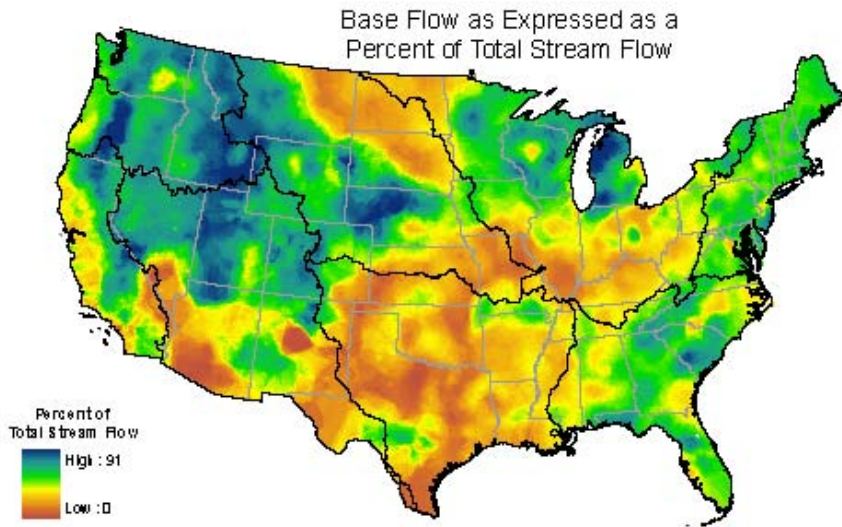
- 1 Channel precipitation
- 2 Overland flow*
- 3 Interflow*
- 4 **Baseflow**

SW/GW Interactions

Baseflow is the component of streamflow that is fed by groundwater and thus maintains flow during rain-free periods. Streams with a large proportion of baseflow tend to have lower variability given the longer residence times of groundwater.

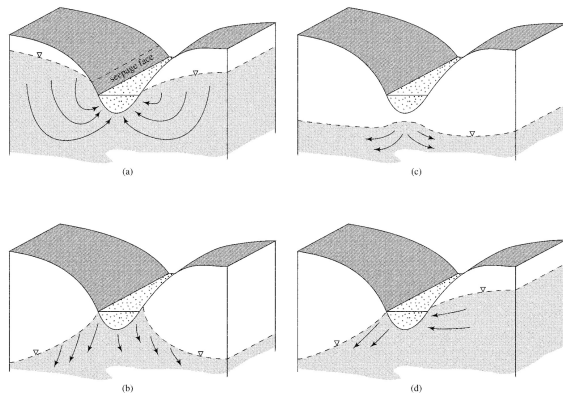


Baseflow Index

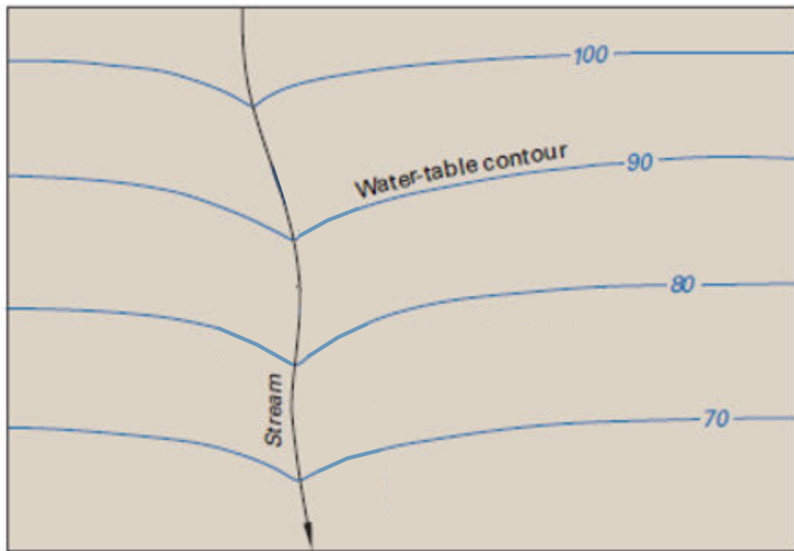


Stream Types

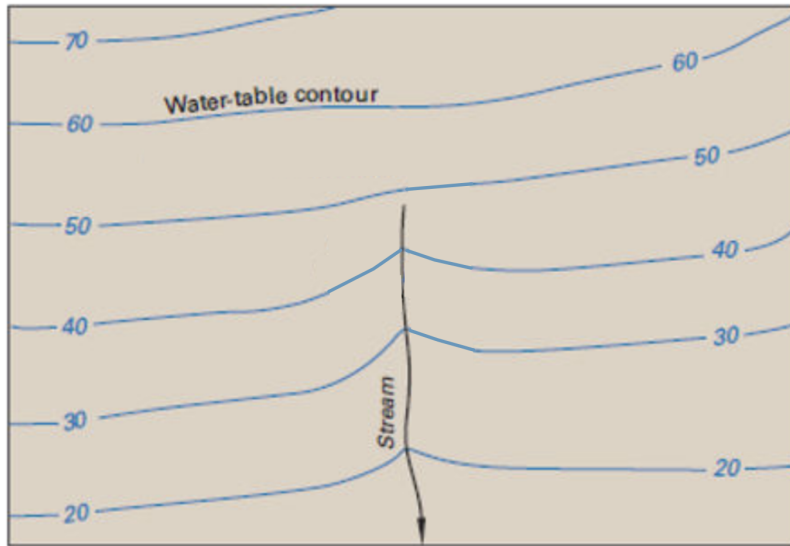
- 1 **Gaining:** WT is higher than stream stage, discharge increases downstream. Typically *perennial*.
- 2 **Losing:** Stream stage higher than WT, discharge decreases downstream. GW mound beneath stream if connected to WT, small mound if not. Typically *ephemeral*.
- 3 **Flow-through:** stream simultaneously gains/loses water.



Guess that stream type!



Guess that stream type!



Guess that stream type!

